Examiner only

2. *(a)* The escape velocity, v, of a mass, m, from a spherical mass, M, and radius, R, can be calculated using:

$$\tfrac{1}{2}mv^2 - \frac{GMm}{R} = 0$$

(i) Explain how this equation is an application of conservation of energy. [3]

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(ii) Calculate the escape velocity from the Sun ($M_{\text{Sun}} = 1.99 \times 10^{30}$ kg, $R_{\text{Sun}} = 6.96 \times 10^{8}$ m). [3]

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(b) (i) The temperature of the surface of the Sun is 5780 K. Use a kinetic theory equation to show that the rms speed of a free electron on the surface of the Sun is approximately 500 km s^{-1}. [4]

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(ii) By considering your answers to *(a)*(ii) and *(b)*(i), explain why the Sun has a slight positive charge. [2]

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(iii) A student claims that a positive charge of approximately 0.08 C on the Sun is enough to produce an electrostatic force equal to the gravitational force on an escaping electron. Determine whether or not she is correct. [3]

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1420U401 07

(iv) Estimate the percentage of lost electrons compared with the total number of electrons on the Sun. Assume that the Sun is mainly hydrogen and that it has lost 0.08 C of charge in the form of electrons. [3]

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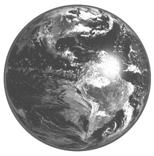
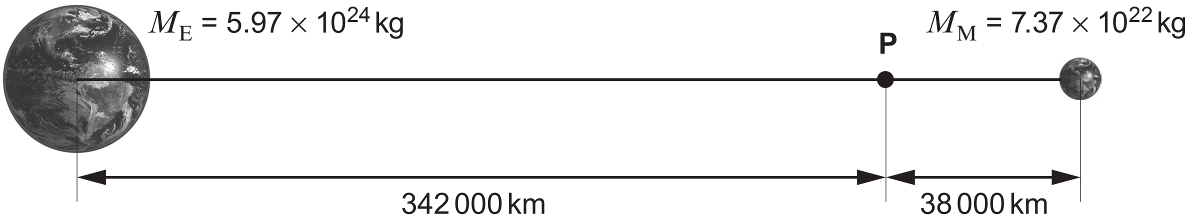
18

Examiner only

2. *(a)* Assuming that the Earth is an isolated perfect sphere, draw its gravitational field lines and equipotential surfaces. [3]



(b) (i) Use the information in the diagram to calculate the gravitational potential at point **P**. [3]



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(ii) Use the information in the diagram to show that the resultant gravitational field at point **P** is very small. [2]

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(iii) Myfanwy correctly calculates that the force on a 25 tonne spaceship would be negligible at point **P** and that the force would increase by approximately 0.5 N for every 10 km moved away from point **P** towards the Earth. Dafydd then concludes that the spaceship will perform simple harmonic motion about point **P**. Deduce whether or not Dafydd is correct **(no further calculations are required)**. [3]

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1420U401 09

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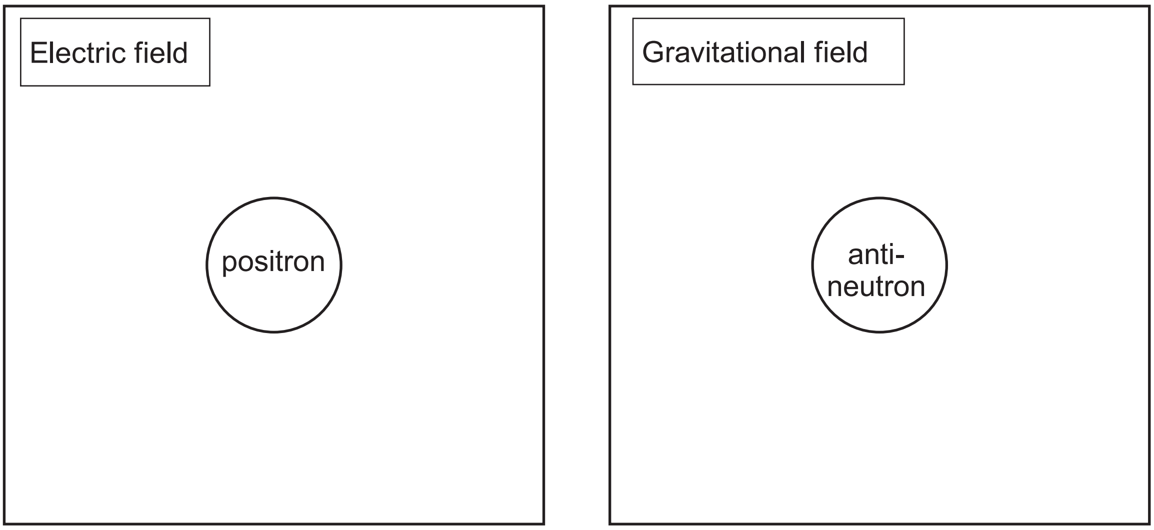
2. (a) Draw 2 equipotentials and 4 field lines for both the **electric** field of the positron and the **gravitational** field of the anti-neutron shown. [3]



(b) Describe and explain the similarities **and** differences between electric **and** gravitational fields. [6 QER]

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9

1420U401 05

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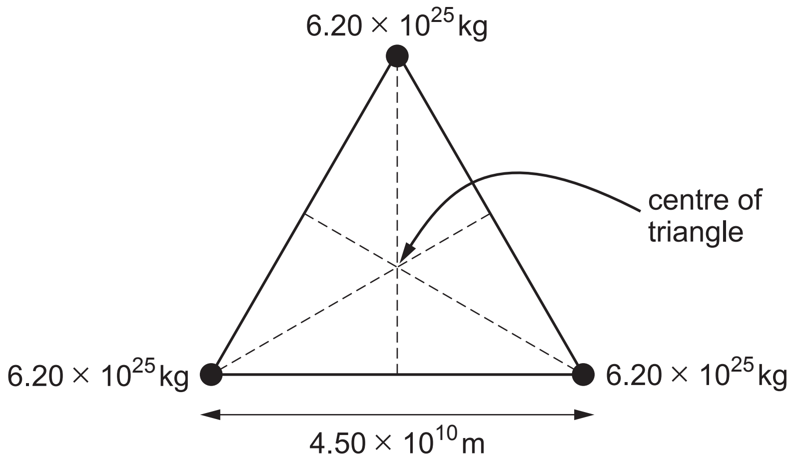
3. Three planets of equal mass are arranged in an **equilateral triangle** as shown.



(a) (i) Show that the gravitational force exerted by one planet on another is approximately 1.3×10^{20} N. [2]

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(ii) Hence, explain why the resultant gravitational force experienced by any of the planets is approximately 2×10^{20} N towards the centre of the triangle. [4]

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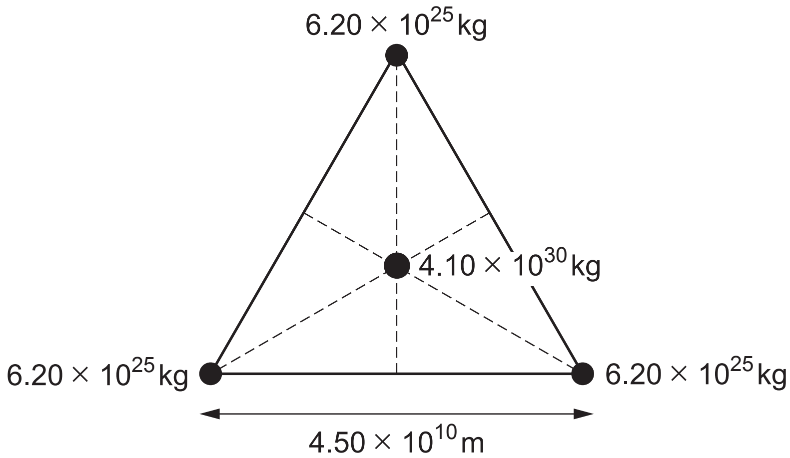
(b) A star is located at the centre of the triangle.



(i) By adding to the diagram, show that the distance between the star and any of the planets is given by: [2]

$$\frac{4.5 \times 10^{10}}{2 \times \cos 30°}$$

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(ii) Rhodri states that the force calculated in part (a)(ii) is negligible compared with the gravitational force exerted on a planet by the star. Determine whether or not Rhodri is correct. [2]

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(c) (i) Show that the gravitational potential at the position of a planet due to the star and the other two planets is approximately $-1 \times 10^{10}\ \mathrm{J\,kg^{-1}}$. [3]

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1420U401
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(ii) Rhodri now states that, if an object with kinetic energy 7×10^{35} J struck one of the planets, the planet would gain enough energy to escape the gravitational pull of the star. Discuss to what extent Rhodri's statement might be correct. [3]

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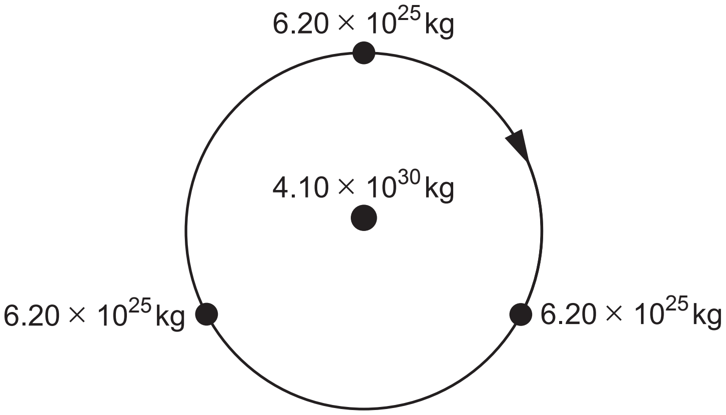
(d) The three planets, whilst orbiting the star, stay in an equilateral triangle. Delyth claims that these planets cannot be detected by the periodic Doppler shift of the star. Determine whether or not Delyth is correct. [3]



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